General Physics IV (Vibration, Waves and Optics) (PHY 104)

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Course Outline

Simple harmonic motion (SHM). Energy in a vibrating system. Damped SHM. Resonance and transient. Coupled SHM. Q values and power response curves. Normal modes. Waves (types and properties of waves as applied to sound). Transverse and longitudinal waves (superposition, interference, dispersion, polarization).

Waves at interfaces (energy and power waves). The wave equation. 2-D and 3-D wave equations. Wave energy and power. Phase and group velocities. Echo and beats. The Doppler-effects. Propagation of sound in gases, solids and liquids and their properties.

Optics: Nature and propagation of light. Reflection and refraction. Internal reflection. Scattering of light. Reflection and refraction at plane and spherical surfaces. Thin lenses and optical instruments. Wave nature of light. Dispersion. Huygens's principle (interference and diffraction).

Simple Harmonic Motion (SHM)

Periodic motion: A motion, which repeat itself at a regular interval of time is called a **periodic motion** and the fixed interval of time after which the motion is repeated is called period of the motion (time period). The path of periodic motion may be linear, circular, elliptical or any other curve. For example, rotation of earth about the sun (period one year), motion of minute's hand of a clock (period 1-hour), etc.

Oscillatory or Vibratory motion: Periodic motion can be **oscillatory** if its to and fro or back and forth about a mean or equilibrium position (e.g. motion of a pendulum bob). Else **non-oscillatory** such as motion of the hands of the clock.

Harmonic and non-harmonic oscillation: Harmonic oscillation is that oscillation which can be expressed in terms of single harmonic function (*i.e.* sine or cosine function). *Example*:
$y = a\sin\omega t$ or $y = a\cos\omega t$

Non-harmonic oscillation is that oscillation which cannot be expressed in terms of single harmonic function. It is a combination of two or more than two harmonic oscillations. *Example:* $y = a\sin\omega t + b\sin 2\omega t$

Some Important Definitions

(1) **Amplitude (a)**: is the maximum displacement of the oscillator on either side of its mean position.

(2) **Time period (T):** It is the least interval of time after which the periodic motion of a body repeats itself.

S.I. units of time period is second.

(3) **Frequency (f):** It is defined as the number of periodic motions executed by body per second. S.I unit of frequency is hertz (*Hz*).

(4) **Angular Frequency (ω):** Angular frequency of a body executing periodic motion is equal to product of frequency of the body with factor 2π. Angular frequency ω = 2

πf

S.I. units of ω is *Hz*. ω also represents angular velocity. In that case unit will be rad/sec.

(5) **Displacement (y or x):** is the distance of the harmonic oscillator from its mean (or equilibrium) position at a given instant.

(6) **Phase (ϕ):** phase of a vibrating particle at any instant is a physical quantity, which completely express the position and direction of motion, of the particle at that instant with respect to its mean position.

In oscillatory motion the phase of a vibrating particle is the argument of *sine* or *cosine* function involved to represent the generalized equation of motion of the vibrating particle.

$y = a\sin\theta = a\sin(\omega t + \phi_0)$ here, $\theta = \omega t + \phi_0$ = phase of vibrating particle.

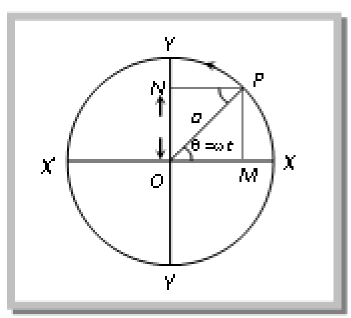
Simple harmonic motion is a special type of periodic motion (oscillatory), in which a particle moves to and fro repeatedly about a mean position under a restoring force which is always directed towards the mean position and whose magnitude at any instant is directly proportional to the displacement of the particle from the mean position at that instant.

Restoring force ∝ Displacement of the particle from mean position.

$$F \propto kx$$

$$F = -kx$$

Where k is known as force constant. Its S.I. unit is Newton/meter.



Displacement of a particle executing S.H.M. is

$y = a\sin\omega t$

$$= a\sin\frac{2\pi}{T}t$$

$$= asin\,2\pi ft$$

$$= a\sin(\omega t \pm \phi)$$

Velocity of the particle executing S.H.M. at any instant, is defined as the time rate of change of its displacement at that instant.

From $y = a\sin\omega t$

$$v = \frac{dy}{dt} = a\omega\cos\omega t$$

$$\therefore \qquad v = a\omega cos\,\omega t$$

Or $\qquad v = a\omega\sqrt{1 - \sin^2\omega t}$

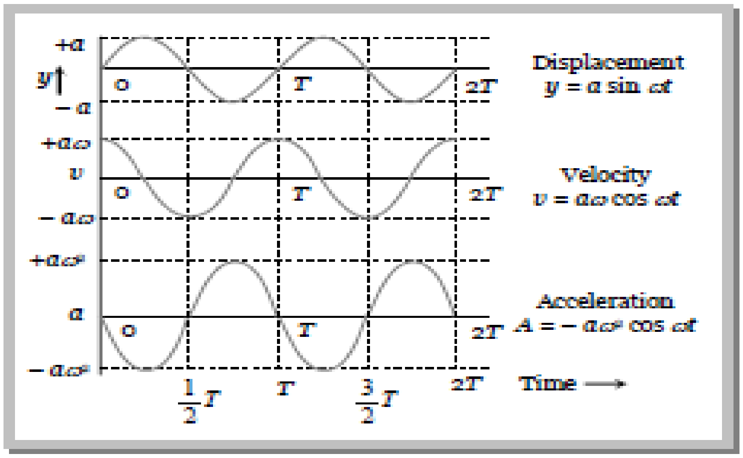
Or $\qquad v = \omega\sqrt{\mathrm{a}^2 - \mathrm{y}^2}$

The acceleration of the particle executing S.H.M. at any instant, is defined as the rate of change of its velocity at that instant. So acceleration,

$$a = \frac{dv}{dt} = -\omega^2 a\sin\omega t$$

$a = -\omega^2 y$ $\qquad$ {where $y = a\sin\omega t$}

Comparing displacement, velocity and acceleration in SHM.



Displacement $y = a\sin\omega t$

Velocity $v = a\omega cos\,\omega t = a\omega\sin(\omega t + \frac{\pi}{2})$

Acceleration $a = -a\omega^2\sin\omega t = a\omega^2\sin(\omega t + \pi)$

Energy in SHM

A particle executing S.H.M. possesses two types of energy: Potential energy and Kinetic energy.

(1) **Potential energy:** This is an account of the displacement of the particle from its mean position.

The restoring force $F = -ky$ against which work has to be done,

$$U = \int dw = -\int_0^x Fdx = \int_0^y kydy = \frac{1}{2}ky^2$$

$\therefore$ Potential energy, $U = \frac{1}{2}m\omega^2y^2$ where $m\omega^2 = k$

$U = \frac{1}{2}m\omega^2a^2\sin^2\omega t$ where $y = a\sin\omega t$

(2) **Kinetic energy:** This is because of the velocity of the particle

Kinetic Energy $K = \frac{1}{2}mv^2$

$K = \frac{1}{2}ma^2\omega^2\cos^2\omega t$ where $v = a\omega cos\,\omega t$

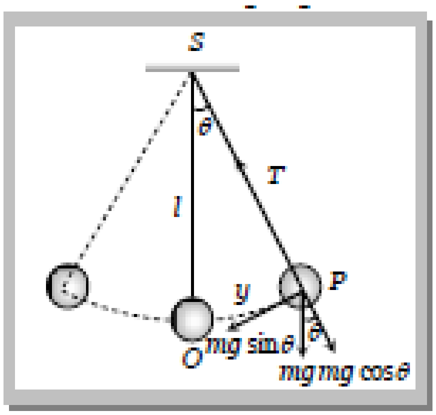
$K = \frac{1}{2} m\omega^2 (a^2 - y^2)$ where $v = \omega\sqrt{a^2 - y^2}$

Total energy : Total mechanical energy = Kinetic energy + Potential energy

$E = \frac{1}{2} m\omega^2 (a^2 - y^2) + \frac{1}{2} m\omega^2 y^2 = \frac{1}{2} m\omega^2 a^2$

Simple Pendulum

An ideal simple pendulum consists of a heavy point mass body suspended by a weightless, inextensible and perfectly flexible string from a rigid support about which it is free to oscillate.



Let mass of the bob = *m*, length of simple pendulum = *l* and displacement of mass from mean position (*OP*) = *x*

When the bob is displaced to position *P*, through a small angle θ from the vertical. Restoring force acting on the bob

$$F = -mg \sin\theta$$

or $F = -mg\theta$ (where θ is small, $\sin\theta \approx \theta = \frac{\text{Arc}}{\text{Length}} = \frac{OP}{l} = \frac{x}{l}$)

or $F = -mg\frac{x}{l}$

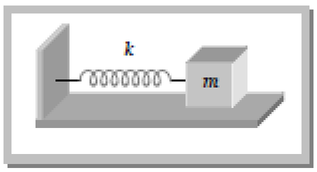
$\therefore$ $\frac{F}{x} = \frac{-mg}{l} = k$ (Spring constant)

So time period, $T = 2\pi\sqrt{\frac{\text{Inertia factor}}{\text{Spring factor}}} = 2\pi\sqrt{\frac{m}{mg/l}} = 2\pi\sqrt{\frac{l}{g}}$

Frequency, $f = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$

Spring Pendulum

A point mass suspended from a mass less spring or placed on a frictionless horizontal plane attached with spring (fig.) constitutes a linear harmonic spring pendulum.



Time period, $T = 2\pi\sqrt{\frac{\text{Inertia factor}}{\text{Spring factor}}} = 2\pi\sqrt{\frac{m}{k}}$ and Frequency, $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

Proof:

For S.H.M. restoring force is proportional to the displacement, $F = -kx$ 1

For S.H.M. acceleration of the body, $a = -\omega^2 y$ 2

$\therefore$ Restoring force on the body, $\therefore F = ma = -m\omega^2 y$ 3

From (1) and (3), $ky = m\omega^2 y$ or $\omega = \sqrt{\frac{k}{m}}$

$\therefore$ Time period, $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$ or Frequency, $f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

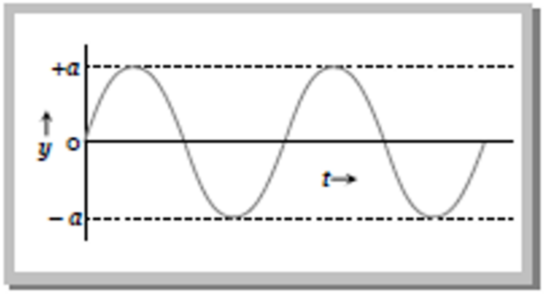
Free, Damped, Forced, Maintained and Coupled Oscillation

(1) Free oscillation

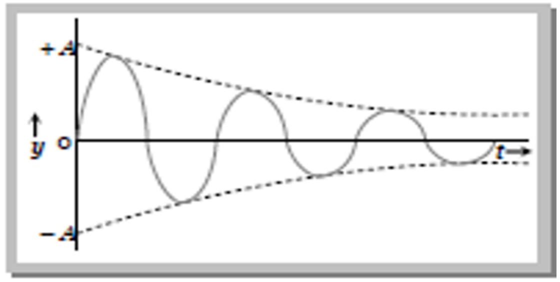
(i) The oscillation of a particle with fundamental frequency under the influence of restoring force are defined as free oscillations

(ii) The amplitude, frequency and energy of oscillation remains constant

(iii) Frequency of free oscillation is called natural frequency because it depends upon the nature and structure of the body.



Free Oscillation



Damping Oscillation

(2) Damped oscillation

(i) The oscillation of a body whose amplitude goes on decreasing with time are defined as damped oscillation

(ii) In this oscillation the amplitude of oscillation decreases exponentially due to damping forces like frictional force, viscous force, hysteresis *etc*.

(iii) Due to decrease in amplitude the energy of the oscillator also goes on decreasing exponentially

(3) Forced oscillation

(i) The oscillation in which a body oscillates under the influence of an external periodic force are known as forced oscillation

(ii) The amplitude of oscillator decreases due to damping forces but on account of the energy gained from the external source it remains constant.

(iii) **Resonance:** When the frequency of external force is equal to the natural frequency of the oscillator. Then this state is known as the state of resonance. And this frequency is known as resonant frequency.

(4) Maintained oscillation

The oscillation in which the loss of oscillator is compensated by the supplying energy from an external source are known as maintained oscillation.

(5) Coupled oscillation

Coupled oscillation is a simple harmonic motion in which two or more particles are linked together and move in a coordinated manner. In this type of motion, the particles are said to be "coupled," and the other particles' motion influences their motion. For example, two masses connected by a spring may exhibit coupled harmonic motion, with one mass moving back and forth in time with the other. The motion of the coupled particles can be described using simple harmonic motion equations, with the coupling between the particles influencing the amplitude, frequency, and other properties of the motion.

Waves

A wave is a transfer of energy through a medium from one point to another. In other words, a wave is a disturbance that carries energy from one point to another without causing any permanent displacement of the medium itself. Some examples of waves include; water waves, sound waves, and radio waves.

If a stone is dropped into a pond or swimming pool, ripples or waves are seen spreading on the surface of the water from the point where the stone was dropped. The water itself does not move in the direction of ripples, but wave transfers energy from one point to another.

There are two major classes of waves: mechanical and electromagnetic waves.

Mechanical waves are waves that require a material medium for their propagation, such as water waves, waves generated in a spring or rope and sound waves.

Electromagnetic waves are waves that do not require a material medium for propagation. Examples are light rays, radio waves, X-rays and gamma rays.

Type of waves

We can distinguish two types of waves.

A wave is said to be a *transverse wave* if the direction of travel of the wave is perpendicular to the direction of vibration of the medium. For example, water waves and waves generated by plucking a string are transverse waves. It can be shown that light and electromagnetic waves are also transverse waves but since no medium is required for their propagation, the vibration cannot be particle vibrations. In these cases the waves are made up of electric and magnetic vibrations of very high frequency.

A wave is said to be *longitudinal* if the direction of travel of the wave is the same as the direction of vibration of the medium, for example, sound waves are longitudinal waves.

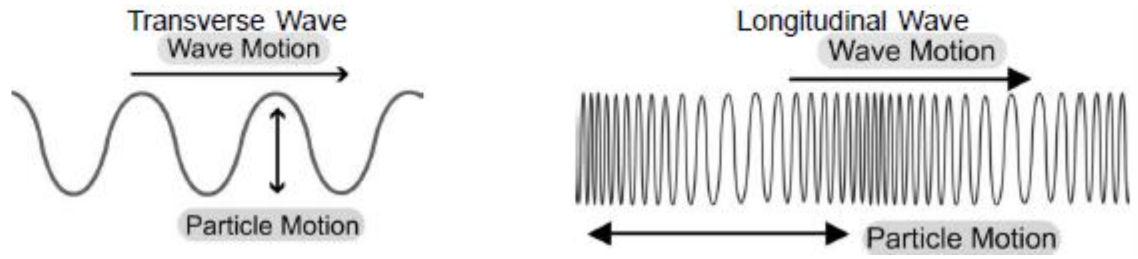


Fig. a

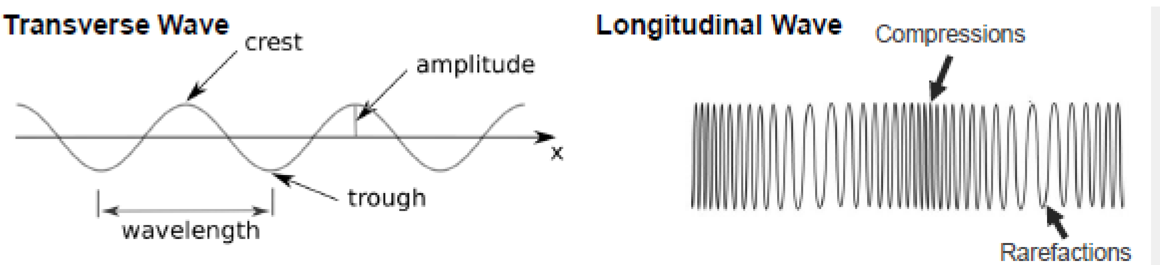


Fig. b.

A transverse wave can be represented pictorially by a series of up and down movements; that is, some portions of the waves are displaced upward while adjacent portions are displaced downwards. The region of maximum upward displacement is called a **crest.** The region of maximum downward displacement is called a **trough.**

The vibrating particles in longitudinal waves behave like a spiral spring that has a series of compressed regions and spaced out regions travelling along it. These are referred to as **compressions** and **rarefactions**.

General representation of wave and definition of terms

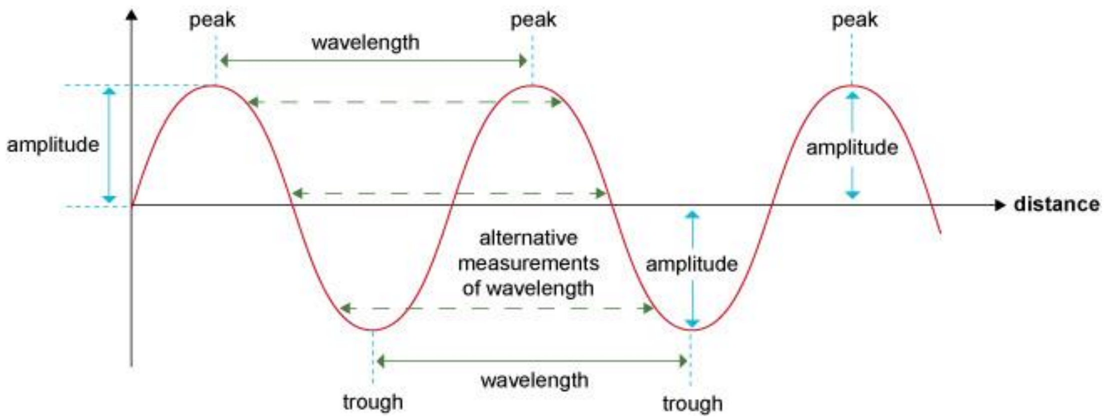


Fig. c. Wave as funtion of distance.

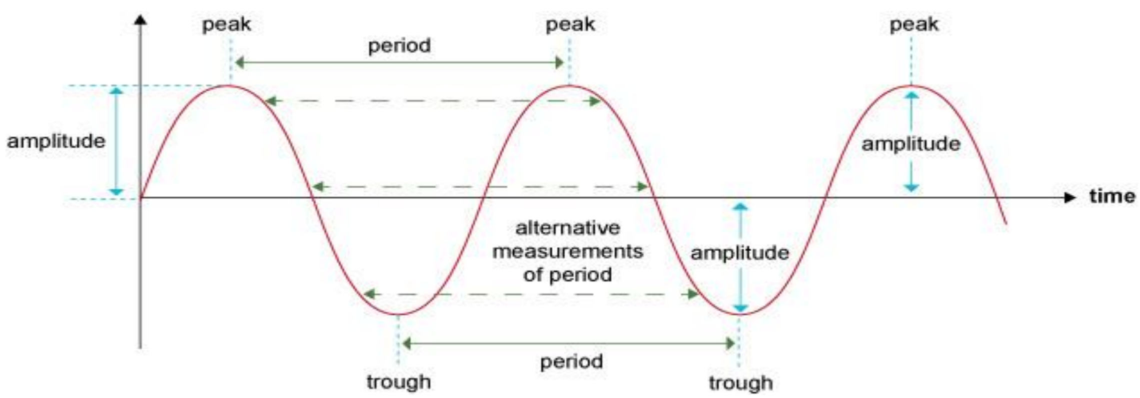


Fig. d. Wave as funtion of time.

1) Amplitude (a): As the wave progresses, the particles of the medium vibrate about a mean position. The maximum displacement of particles from their mean (or rest) position is called the amplitude of the wave. It is measured in meters.

2) Period (T): The time required for a particle to perform one complete cycle or to complete an oscillation is called the period T of the wave. It is measured in seconds. It is also the time taken to cover one wavelength.

3) Frequency (f): It is defined as the number of cycles performed by the wave per second. It is measured in Hertz (Hz),

4) *Wavelength (λ):* is the distance between the successive crest or trough, that is to say a distance covered for one cycle. It is measured in meter (m).

5) *Wave speed (v):* is the distance travelled by the wave per second

The frequency is the reciprocal of period:

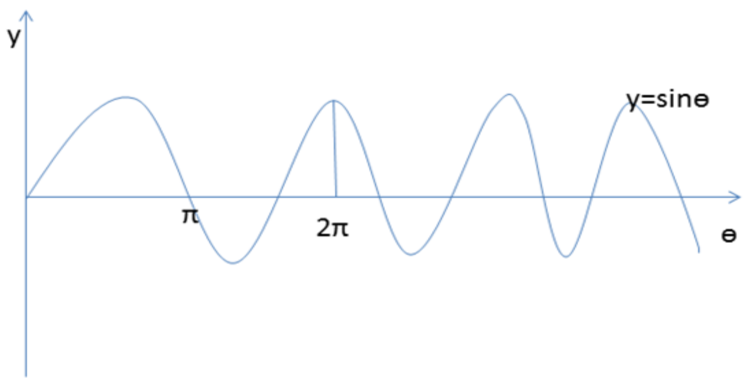
$$f = \frac{1}{T}$$

Also wave speed, $v = \frac{x}{t} = \frac{\lambda}{T} = f\lambda$

$$\therefore \qquad v = f\lambda$$

General equation of a travelling wave

The equation for a travelling wave is written in either sine or cosine function



The wave is represented as a periodic function with amplitude equal to one and period equal to 2π radians or 360°. For a circular motion we know that angular velocity $\omega = \frac{\theta}{t} = \frac{2\pi}{T}$ or $\theta = \omega t$.

Therefore, the equation for the travelling wave whose amplitude is A in a sine function can be written as

$$y = A\sin\omega t$$

For the wave that does not start from the origin,

$$y = A\sin(\omega t - \phi)$$

where ϕ is the angular distance called phase constant, $\phi = \frac{2\pi x}{\lambda}$

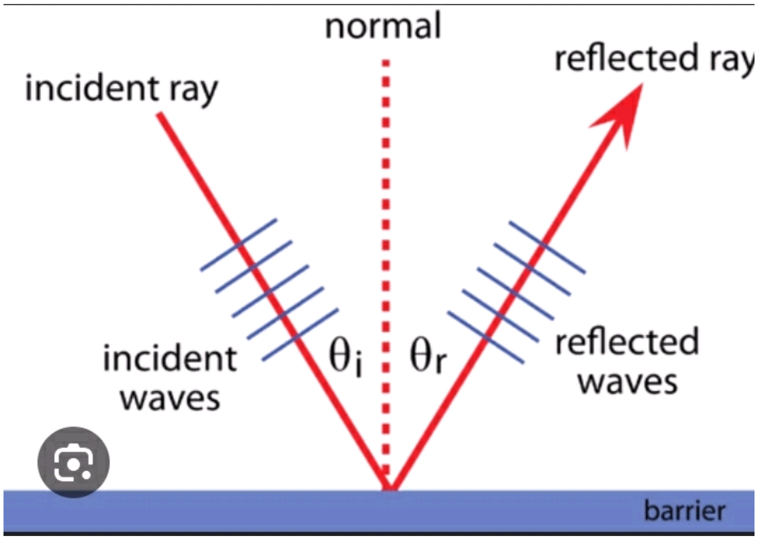
and $k = \frac{2\pi}{\lambda}$ is called wave number.

$$\therefore \qquad y = A\sin(\omega t - kx)$$

Properties of Waves

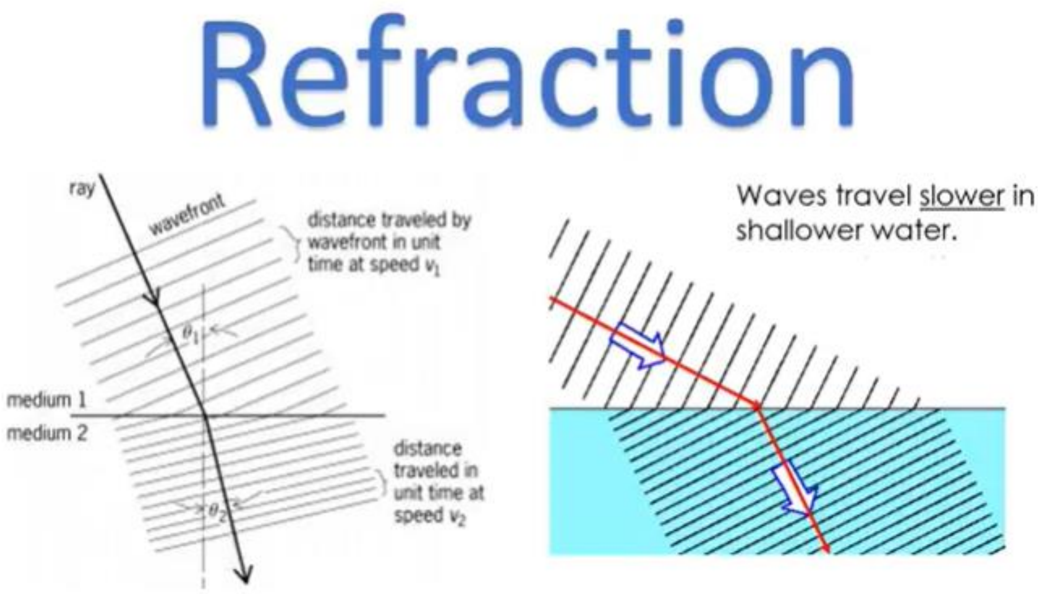
1. Reflection 2. Refraction 3. Diffraction 4. Interference 5. Polarization

1) Reflection: This is a property of wave which occurs when a travelling wave strikes a surface and it bounces back. The travelling wave is the incident wave while the one that bounces back is the reflected wave. In case of water waves generated in a ripple tank, if the waves were made to incident normally on a plane strip, the wave will be reflected back along their original course.



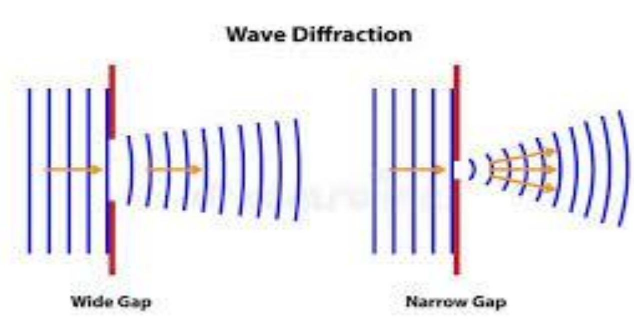
If the waves are incident at a particular angle (flat or plane surface), it will be observed that the angle of incidence is equal to the angle of reflection in line with the laws of reflection. That is, The incident ray, the reflected ray and the normal, at point of incidence, all lie on the same plane. The angle of incidences is equal to the angle of reflection.

2) Refraction: This is the change in the speed and direction of waves as it passes from one medium to another. When plane waves pass from deep to shallow water, their wavelength becomes shorter and thereby travels slowly. A change in the wavelength and speed produce a change in the direction of travel of waves when they cross the boundary. It is important to note that during refraction, the frequency remains constant.



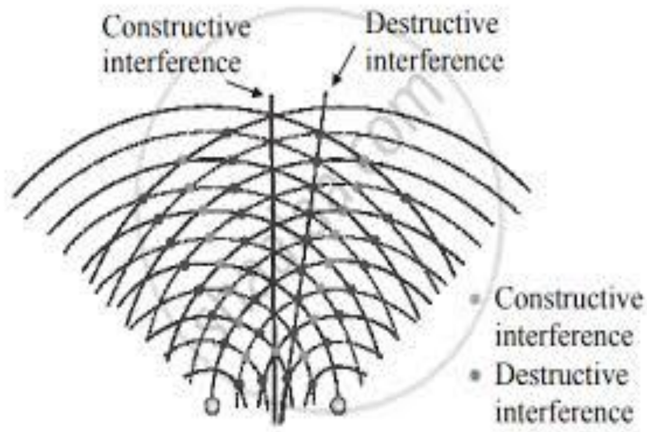
Refractive index is the ratio of the sine of the incident angle (i) to the sine of the angle of refraction (r). It is also the ratio of the velocity of the wave in the first medium (v_1) to the velocity in the second medium (v_2).

3) Diffraction: This is the spreading out of a wave on passing through a narrow opening. If waves are directed towards a large gap compared with the wavelength of the waves, slightly bent or beam of waves are formed on passing through the gap. If the barriers are placed closer to leave a narrow gap waves forms spherical wave fronts on passing through a narrow shit.



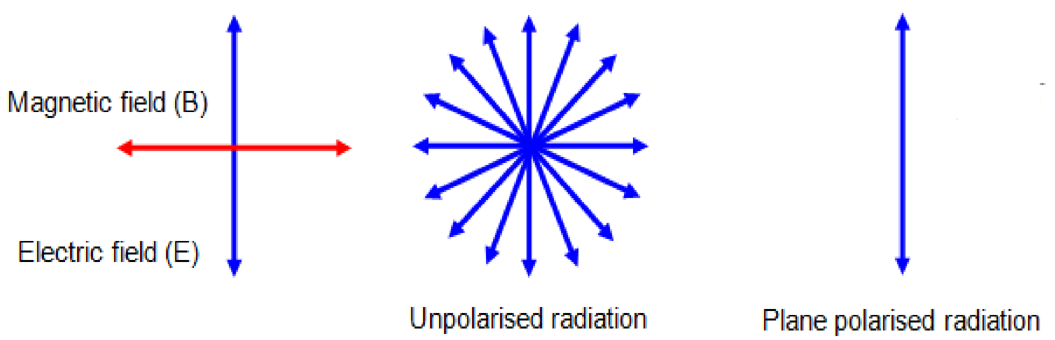
Diffraction occurs when the wavelength of the wave is longer than the width of the opening or the size of the obstacles.

4) Interference: This is a phenomenon which occurs when two similar waves traveling in the same direction cross each other. If the waves are in phase or step so that they travel the same distance at equal time and the crest or trough of the two waves arrive simultaneously or one is a complete wavelength ahead of the other. The resulting wave will build up to twice the amplitude of the two waves; this is called constructive or additive interference.

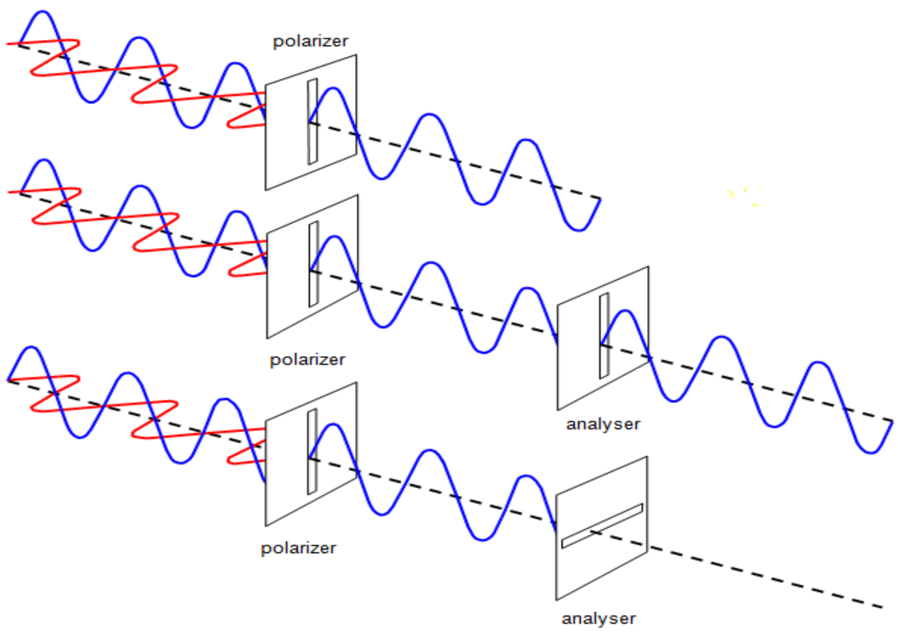


If the crest of one wave arrives with the troughs of the waves, and vice-versa, the waves cancel each other out to give zero resultant, this is called destructive interference.

5) Polarisation: This is an exclusive property of transverse waves only. It is the production of transverse vibration in only one plane. A transverse wave which vibrates in only one plane is said to be plane-polarized.



Polarized light can be produced by passing an ordinary light through a polarizer called Polaroid or crystal of calcite, tourmaline or quartz. The arrangement of molecules within this polarizer will only permit the passage of light in a particular plane and then absorb light due to other vibration. Thus, when an unpolarized light is passed through a polarizer, the emergent light consists in only one plane.



Examples

SHM

1. A particle of mass 200 g executes a simple harmonic motion. The restoring force is provided by a spring of spring constant 80 N/m. Find the time period.

Solution: $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{200\text{x}10^{-3}}{80}} = 0.31\ s$

2. A particle of mass 0.50 kg executes a simple harmonic motion under a force F = - (50 N/m)x. If it crosses the centre of oscillation with a speed of 10 m/s, find the amplitude of the motion.

Solution: KE at the center of ocsillation, $KE = \frac{1}{2}mv^2 = \frac{1}{2}(0.50)(10)^2 = 25$ J

The potential energy is zero here. At the maximum displacement x = A, v = 0 and hence KE = 0.

Since no loss of energy, $PE = \frac{1}{2}kA^2 = 25$ J

and given, $F = -50x,$ where $k = 50$ N/m

$\frac{1}{2}(50)A^2 = 25$ J, $\therefore$ $A = 1$ m

3. A spring of force constant 1600 N m^{-1} is mounted on a horizontal table. A mass m = 4.0 kg attached to the free end of thespring is pulled horizontally towards the right through a distance of 4.0 cm and then set free. Calculate (i) the frequency (ii) maximum acceleration and (iii) maximum speed of the mass.

Solution: i) $\omega = \sqrt{\frac{1600}{4}} = 20$ rad s^{-1}

$$f = \frac{20}{2\pi} = 3.18 \text{ Hz}$$

ii) Maximum acceleration, $a = \omega^2 y = (20)^2(0.04) = 16$ m s^{-1}

iii) Max speed, $v = \omega y = 20\text{x}0.04 = 0.8$ m s^{-1}

Wave

1. A travelling wave in a string is given by y = 0.5sin (52t−3x). Find the amplitude, wavelength, frequency, the period and speed of the wave.

Solution: Amplitude, A = 0.5 m,

Wavelength, $\lambda = \frac{2\pi}{k} = \frac{2\pi}{3} = 2.1$ m

Frequency, $f = \frac{\omega}{2\pi} = \frac{52}{2\pi} = 7.6$ Hz

Period, $T = \frac{1}{f} = \frac{1}{7.6} = 0.13$ s

Speed, $v = f\lambda$ = 7.6x2.1 = 15.96 m s^{-1}